

Name
Date
Course

Evaluating Composite Trigonometric Functions (Part II) Homework

Find the value of each expression *without a calculator*.

① $\sec^{-1}(\csc 120^\circ)$

② $\cot^{-1}(\sec 180^\circ)$

$$\textcircled{3} \cos^{-1}(\tan(-225^\circ))$$

$$\textcircled{4} \sin^{-1}(\cos 330^\circ)$$

$$\textcircled{5} \cos^{-1}(\cos 45^\circ)$$

$$\textcircled{6} \csc^{-1}(\csc 30^\circ)$$

$$\textcircled{7} \cot^{-1}(\cot(-90^\circ))$$

$$\textcircled{8} \sin^{-1}(\sin 150^\circ)$$

$$\textcircled{9} \sec^{-1}(\sec 240^\circ)$$

$$\textcircled{10} \tan^{-1}(\tan 135^\circ)$$

$$\textcircled{11} \sin^{-1}\left(\cos\left(-\frac{5\pi}{6}\right)\right)$$

$$\textcircled{12} \cot^{-1}\left(\tan\frac{\pi}{3}\right)$$

$$\textcircled{13} \cos^{-1}\left(\cos\frac{5\pi}{6}\right)$$

$$\textcircled{14} \cot^{-1}\left(\cot\frac{3\pi}{4}\right)$$

$$\textcircled{15} \sin^{-1}\left(\sin \frac{3\pi}{2}\right)$$

$$\textcircled{16} \sec^{-1}\left(\sec \frac{5\pi}{3}\right)$$

Homework Answers

$$\textcircled{1} \boxed{30^\circ}$$

$$\textcircled{12} \boxed{\frac{\pi}{6}}$$

$$\textcircled{2} \boxed{135^\circ}$$

$$\textcircled{13} \boxed{\frac{5\pi}{6}}$$

$$\textcircled{3} \boxed{180^\circ}$$

$$\textcircled{14} \boxed{\frac{3\pi}{4}}$$

$$\textcircled{4} \boxed{60^\circ}$$

$$\textcircled{15} \boxed{-\frac{\pi}{4}}$$

$$\textcircled{5} \boxed{45^\circ}$$

$$\textcircled{16} \boxed{\frac{\pi}{3}}$$

$$\textcircled{6} \boxed{30^\circ}$$

$$\textcircled{7} \boxed{90^\circ}$$

$$\textcircled{8} \boxed{30^\circ}$$

$$\textcircled{9} \boxed{120^\circ}$$

$$\textcircled{10} \boxed{-45^\circ}$$

$$\textcircled{11} \boxed{-\frac{\pi}{3}}$$